

# Anirban Ray

Final-year PhD Student in Computer Science

Computational Image Restoration | Generative AI for Microscopy

Email | [Personal Page](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#)

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## Profile

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Final-year Computer Science PhD student developing generative AI and deep learning methods for computational microscopy. Research focus: inverse problems, image restoration, dehazing, super-resolution, denoising, uncertainty estimation, and posterior sampling. Prior industry experience as an AI researcher at Hitachi R&D in Tokyo.

## Education

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### PhD in Computer Science

2022–2026 (expected)

*Technische Universität Dresden & Human Technopole*

*Dresden, Germany / Milan, Italy*

- Advisor: Dr. Florian Jug; Thesis Advisory Committee: Prof. Dr. Ivo F. Sbalzarini and Dr. Virginie Uhlmann.
- Thesis: *Generative Image Restoration for Microscopy*.
- Selected outputs: ResMatching (IEEE ISBI 2026 oral), HazeMatching (CVPR 2026 Findings), and RESOLFT deep-learning imaging (Nature Communications).

### Master of Engineering in Computer Science

2016–2018

*Nagoya Institute of Technology*

*Nagoya, Japan*

- Advisors: Prof. Jun Sato and Prof. Fumihiko Sakaue.
- Thesis: *Modeling the Feature Evolution in CNNs using LSTM*; focus on computer vision and deep learning.

### Research Student

Oct. 2015–Mar. 2016

*Nagoya Institute of Technology*

*Nagoya, Japan*

### Bachelor of Technology in Computer Science and Engineering

2011–2015

*Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology*

*Chennai, India*

## Research Experience

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### RESOLFT time-lapse imaging empowered by deep learning

2024–Present

*Human Technopole; collaboration with the Testa Lab*

*Milan, Italy*

- Applied deep learning restoration to low-SNR, sub-sampled RESOLFT nanoscopy, enabling  $5\times$  longer imaging with  $10\times$  lower light dose or  $4\times$  faster imaging while preserving approximately 60 nm resolution.
- Output: Nature Communications, 2026 (in Press); [Research Square preprint](#).

### ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching

2024–2025

*Human Technopole*

*Milan, Italy*

- Developed guided conditional flow matching for microscopy super-resolution, unifying denoising, uncertainty estimation, and posterior sampling.
- IEEE ISBI 2026 oral presentation; [GitHub](#), [arXiv](#), [project page](#).

### HazeMatching: Conditional Flow Matching for Microscopy Dehazing

2023–2024

*Human Technopole*

*Milan, Italy*

- Created a conditional flow-matching framework for microscopy dehazing, widefield-to-confocal restoration, and Expansion Microscopy super-resolution.
- CVPR 2026 Findings; CV4Science Workshop poster; [GitHub](#), [arXiv](#).

**Deep Learning for Microscopy Image Analysis** **2018–2021**  
*Hitachi Ltd., Center for Technology Innovation* *Tokyo, Japan*  

- Built deep learning systems for biomedical microscopy and SEM image analysis, including segmentation, counting, classification, quantification, CISS publications, and granted patents.

**Modeling the Feature Evolution in CNNs using LSTM** **2016–2018**  
*Nagoya Institute of Technology* *Nagoya, Japan*  

- Modeled feature evolution across CNN layers using LSTM networks for image classification.

## Professional Experience

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**Associate Researcher** **Apr. 2018–Jan. 2022**  
*Hitachi Ltd., R&D Group, Center for Technology Innovation* *Tokyo, Japan*  

- Delivered deep learning and computer vision systems for industrial R&D and biomedical microscopy image analysis.

**Engineering Intern, AR Applications** **Sept. 2016**  
*Sun Corporation* *Konan, Japan*

**Engineering Intern, Cloud Services** **Aug. 2014–Jan. 2015**  
*Machine Pulse (Mahindra Tego)* *Mumbai, India*

## Publications and Preprints

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- [1] **Anirban Ray et al.**. [ResMatching: Noise-Resilient Computational Super-Resolution via Guided Conditional Flow Matching](#). *IEEE ISBI 2026*. Accepted; oral presentation.
- [2] **Anirban Ray et al.**. [HazeMatching: Conditional Flow Matching for Microscopy Dehazing](#). *CVPR 2026 (Findings)*. Accepted; CV4Science Workshop poster.
- [3] **Minet, Ray et al.**. [RESOLFT time lapse imaging empowered by deep learning](#). *Nature Communications 2026 (in Press)*.
- [4] **Kakishita, Ray et al.**. [Quantitative Analysis System for Bacterial Cells in SEM Image using Deep Learning](#). *CISS 2021*. pp. 1–6.
- [5] **Kakishita, Ray et al.**. [Deep Learning Based Bacteria Classification from SEM Images Using a Combination of Membrane and Internal Features](#). *CISS 2022*. pp. 13–18.

## Patents

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- Inventor on 3 granted Hitachi High-Tech image-processing patents: [US12327363B2](#), [US12211213B2](#), and [EP3961562B1](#).

## Presentations and Posters

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- **Invited talk:** Scientific Inverse Problems in the Age of Generative AI. AI & Advanced Computing Institute, Schmidt Sciences, New York, USA, Jun. 2026.
- **Talk and poster:** [ResMatching](#). IEEE ISBI 2026, London, UK, Apr. 2026.
- **Poster:** [HazeMatching](#). IEEE/CVF CVPR 2026 and CV4Science Workshop, Denver, CO, USA, Jun. 2026.
- **Invited talk:** Introduction to Diffusion Model for Microscopy. EMBO-DL4MIA, Milan, Italy, May 2024.

## Teaching and Mentoring

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**Teaching Assistant, Deep Learning for Microscopy Image Analysis (DL4MIA)** **2022, 2023, 2024**  
*Human Technopole* *Milan, Italy*  

- Mentored participants, supported coding sessions and projects, and delivered an invited lecture in the 2024 edition.

## Teaching Assistant, Deep Learning at MBL (DL@MBL)

2023, 2024

Marine Biological Laboratory

Woods Hole, USA

- Led hands-on support for deep learning in biological imaging; co-mentored project work linked to NeurIPS 2024 and RESOLFT microscopy outputs.

## Professional Service

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- Reviewer, [AI4Life Open Call](#), 2024.

## Awards, Honors, and Selected Training

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- **Selected Participant, Optical Microscopy & Imaging in the Biomedical Sciences (OMIBS)**, Marine Biological Laboratory, USA, 2023; team won 3rd place in microscopy imaging.
- **Selected Participant, International Computer Vision Summer School (ICVSS)**, Italy, 2017; selected among 150 global students.
- **Aichi Monozukuri Scholarship**, Aichi Prefectural Government, Japan, 2015; full funding for master's studies, selected among top 10 students in Asia.
- **Selected Participant, Undergraduate Summer School on Computer Science**, Indian Institute of Science, Bengaluru, India, 2013.

## Projects and Software

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- **ResMatching**: Open-source implementation, data, models, project page, and quickstart notebook for guided conditional flow matching in microscopy super-resolution.
- **HazeMatching**: Open-source implementation, data, models, and quickstart notebook for conditional flow matching in microscopy dehazing and Expansion Microscopy super-resolution.

## Skills

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|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research</b>              | Computational microscopy; image restoration; inverse problems; generative AI; conditional flow matching; diffusion models; denoising; dehazing; super-resolution; uncertainty estimation; posterior sampling; segmentation; computer vision. |
| <b>Programming Languages</b> | Python; PyTorch; OpenCV; NumPy; Bash; Linux; Git; SLURM; $\text{\LaTeX}$ .<br>Bengali (native); Hindi (fluent); English (fluent); Japanese (conversational); Tamil (elementary); Italian (elementary).                                       |